

JONATHAN LUKWICHI

MSc Candidate in Industrial Engineering | AI & Machine Learning Practitioner | Supply Chain Analyst

BSc Hons Industrial Systems | BTech Electromechanical Engineering

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Hillcrest, Pretoria, South Africa | Open to Relocation

PROFESSIONAL PROFILE

MSc Candidate in Industrial Engineering and AI & Machine Learning Practitioner with hands-on experience building intelligent forecasting systems and end-to-end ML pipelines. Passionate about applying AI and machine learning to transform traditional supply chain operations, replacing reactive, manual approaches with smart, data-driven solutions that deliver real efficiency and cost impact.

Solid engineering foundation across electrical systems, industrial automation, and supply chain optimisation, enabling the delivery of AI solutions that are technically sound, practically grounded, and commercially relevant for African and emerging-market contexts.

CORE TECHNICAL COMPETENCIES

- Artificial Intelligence & Machine Learning
- Time Series Forecasting & Demand Planning
- Python Programming (Pandas, NumPy, Scikit-learn, Matplotlib)
- Data Science & Exploratory Data Analysis
- Statistical Analysis & Predictive Modelling
- Supply Chain Analytics & Optimisation
- Azure Machine Learning & Cloud AI Platforms
- Prompt Engineering & Large Language Models
- Data Visualisation (Power BI, Excel, Seaborn)
- ETL Processes & Data Pipeline Design
- Process Optimisation (Six Sigma Green Belt, Lean Manufacturing)
- Industrial Automation & Control Systems
- CAD Design (AutoCAD, SolidWorks, Autodesk Inventor)
- Electrical & Electronic Systems Engineering

PROFESSIONAL EXPERIENCE

Teaching Assistant, Data Analytics & Engineering | 2025 Feb – Dec 2025

University of Pretoria | Dept. of Mechanical & Aeronautical Engineering

- Provide technical instruction to 200+ engineering students across Graphical Communication, Professional Communication, and Engineering Impact modules, applying data analysis to monitor and continuously improve learning outcomes
- Leverage Python (Pandas, NumPy) and Excel to analyse student performance trends, enabling evidence-based adjustments to teaching methodologies and assessment design
- Contribute to academic research in AI-driven forecasting and predictive analytics as part of the MSc Industrial Engineering research programme

Customer Service Representative | Feb 2025 – May 2025

ACT TODAY Call Center

- Managed high-volume client interactions in a fast-paced environment, developing advanced communication, prioritisation, and problem-solving capabilities directly transferable to client-facing analytics and consulting roles

Engineering Intern, Electrical & Process Engineering | Jan 2019 – Mar 2019

Gécamines (State Mining Company), Lubumbashi, DRC

- Maintained and optimised electrical power distribution systems, electronic equipment, and mechanical machinery across large-scale industrial mining operations, building hands-on expertise in mission-critical engineering environments

- Programmed and configured microcontrollers for automated process control, directly contributing to manufacturing process optimisation and measurable uptime improvement
- Designed and implemented data-driven safety inspection strategies to ensure operational reliability and full compliance with mining industry standards

Instrumentation Intern | Jul 2013 – Aug 2013

Ruashi Mining, Lubumbashi, DRC

- Supported installation, calibration, and commissioning of industrial instrumentation systems in mineral processing facilities, ensuring measurement accuracy and operational reliability
- Conducted systematic preventive maintenance and troubleshooting to minimise production downtime and maintain measurement integrity across processing lines

EDUCATION

Master of Science (MSc), Industrial Engineering [Candidate] | 2024 – Present

University of Pretoria

- Thesis Research: Optimizing Hospital Supply Chain Demand Forecasting Using Machine Learning: A Case Study of South African Public Hospitals

Bachelor of Science Honours (BSc Hons), Applied Science in Industrial Systems | 2022 – 2023

University of Pretoria

- Key Modules: Supply Chain Design, Modelling & Simulation, Industrial Analysis, Manufacturing Control & Planning, Operations Research.

Bachelor of Engineering Technology (BEng Tech), Electromechanical Engineering | 2014 – 2019

University of Lubumbashi, DRC

- Key Modules: Electrical Circuits, Electronics, Mechanical Systems Design, Computer-Aided Design (CAD), Simulation Software

National Senior Certificate, Industrial Electronics | 2008 – 2014

Salama Technical Institute, DRC

PROFESSIONAL CERTIFICATIONS & TRAINING

AI & Machine Learning

- **The Complete Azure Machine Learning Course, 2026 Edition** | Udemy
- **ChatGPT for Python Data Science and Machine Learning** | Udemy
- **RA: Data Science and Supply Chain Analytics A-Z with Python** | Udemy
- **No-Code Machine Learning: Practical Guide to Modern ML Tools** | Udemy
- **Modern Artificial Intelligence with Zero Coding** | Udemy
- **Machine Learning Applications** | Udemy

Prompt Engineering

- **Advanced Prompt Engineering Certification** | Udemy
- **Prompt Engineering Certification: Master AI** | Udemy

Industrial Engineering, Supply Chain & Business Analytics

- **Certified AI Supply Chain Analyst (CAISCA)**, YouAccel Training | Udemy
- **Strategic Supply Chain Trade-Offs and Optimisation Using AI** | Udemy
- **CSCMP Supply Chain Foundations: Demand Planning Professional Certificate, CSCMP** | LinkedIn Learning
- **Excel Data Analysis for Supply Chain: Forecasting** | LinkedIn Learning

Programming

- **Python for Everybody**, University of Michigan | Coursera

Quality Management

- **Six Sigma Green Belt**, Udemy | *Process improvement, DMAIC methodology, and quality management*
- **Faculty Teaching Assistant Training**, University of Pretoria | *Pedagogical methodology, assessment design, and student support*
- **Customer Service Representative Level 1 & 2**, ACT TODAY | *Professional communication, client relations, and service excellence*

LANGUAGES

- English, Professional Full Working Proficiency
- French, Native / Bilingual Proficiency

References Available Upon Request